Freshness and Independent Supervision

An operating-systems study on a heterogeneous edge rig

Raspberry Pi 5 / Ubuntu 24.04 + STM32F446RE / FreeRTOS

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Defense evidence snapshot: 13 October 2026

1. Motivation

Artificial intelligence (AI) is a variable Linux workload.

- A late result can be correct but unusable.
- A live process can keep sending stale results.
- A separate supervisor can reject stale proposals.

Figure 1 locates the consumer's freshness boundary.

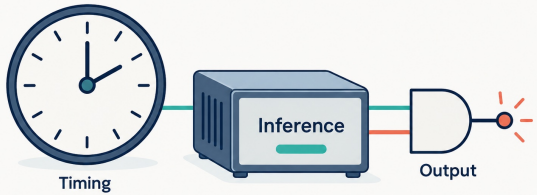


Figure 1: The consumer needs timely information.

Study when information remains useful, alongside how fast work completes.

2. Problem and research questions

- **RQ1:** How does contention change Linux timing tails?
- **RQ2:** Where should a latest-value policy discard old work?
- **RQ3:** Can the microcontroller (MCU) distinguish silence from stale results?

Metrics: wake-up latency, input-to-reception delay,
Age of Information (AoI), and reason-coded state transitions.

> The contribution is an OS experiment and evidence trail; AI supplies load.

3. System architecture

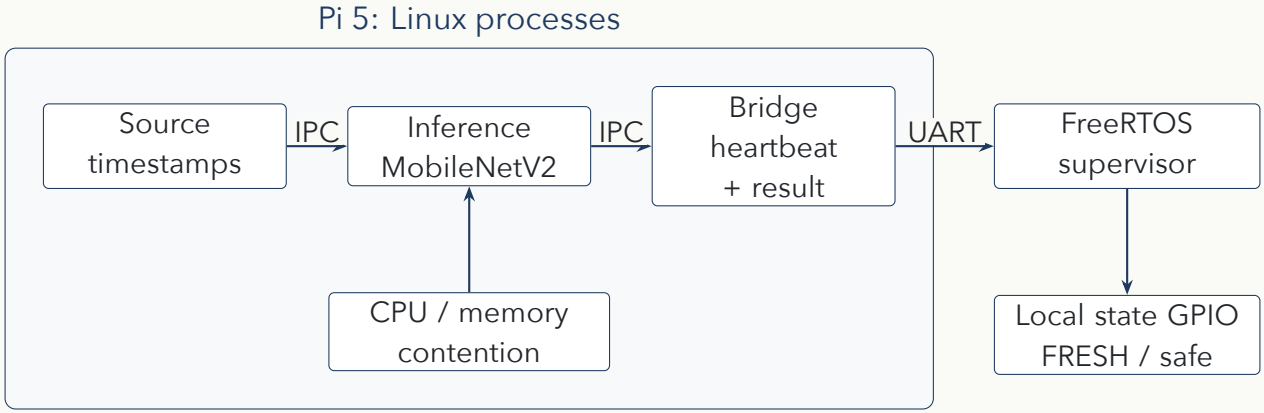


Figure 2: A UART boundary separates Linux work from MCU judgment.

Figure 2 separates inter-process communication (IPC) from the UART serial link.

The STM32 supervises freshness independently of Linux scheduling.

4. Scheduling and contention profiles

Applied software-campaign profiles

- P0 source/bridge: OTHER, CPU 1, unlocked.
- P1 source/bridge: FIFO 60/70, CPU 1, memory locked.
- Both profiles: inference OTHER, CPU 0, unlocked.

Mechanisms and scope

CPU (central processing unit) affinity controls placement.

`mlockall` locks resident memory.

Control groups (cgroups) can cap CPU service.

PLANNED: quota and packaged P3 kernel comparisons; no measured benefit.

[1] Linux scheduling. Applied E2 settings change priority and locking together. Cyclictest P1 is FIFO 50, a separate configuration. Loads and stressed cores: method, page 9.

> CPU priority and affinity do not isolate shared memory bandwidth.

5. IPC and buffering policy

Unix datagram sockets separate source, inference and bridge processes.



Figure 3: Replacement acts only at its chosen waiting queue.

Figure 3: bridge FIFO holds up to 256 results; latest keeps one unsent result.

Input latest drains waiting inputs before inference; it does not preempt current work.

Replacing after expensive work cannot remove a backlog before that work.

6. Clocks and freshness

One Pi clock per measurement

`CLOCK_MONOTONIC`: release deadlines.

`CLOCK_MONOTONIC_RAW`: recorded timestamps and age-at-send.

Do not mix those timestamp domains.

[2] Linux clock semantics.

Independent MCU clock

$A_s = t_{send} - t_{input}$ on the Pi.

$\widehat{A} = A_s + (t_{MCU} - t_{receipt}) + B$.

$B = 5$ ms: configured transit allowance.

Echo exchange fits offset and drift [3].

Mapping uncertainty remains; no proven transit bound.

AoI is the age of the newest useful input at the consumer.

7. Independent supervisor

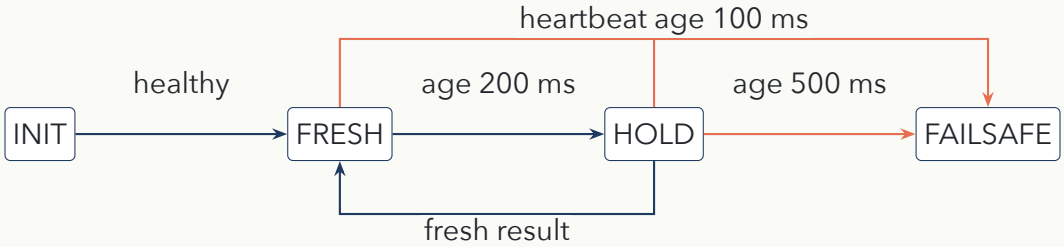


Figure 4: Separate timers test liveness and source freshness.

Figure 4 uses configured, provisional thresholds.

FAILSAFE is latched. Portable core supports local healthy rearm (1 s); this port uses board reset.

FreeRTOS fixed-priority tasks [4]; single supervisor owner, static resources. No independent MCU hardware watchdog.

> A continuing heartbeat cannot make an old result fresh.

8. Experimental method

Retained rig

Pi 5, 4 GB; Ubuntu 24.04.
Stock `6.8.0-1064-raspi` kernel.

STM32F446RE; FreeRTOS.
USART1, 921600 baud; direct memory access (DMA) reception.

3.3 V UART: Pi TX/RX ↔ PA10/PA9; shared ground.

Load key (stress-ng on the Pi)

CPU: compute workers, CPUs 2/3.
Stream: streaming arrays, CPU 2.
Cache: 2 MiB working set, CPU 2.
Memrate: read/write traffic, CPUs 2/3.

Timer: CPU 0; pipeline inference: CPU 0; source/bridge: CPU 1.

Cyclictest: 3×120 s/cell; E1–E3: 3×60 s/cell, pseudo-terminal (PTY).
Inference: 50 warm-ups + 1,000 calls/run; analyzer: 15 fault captures.

Performance governor: timer/pipeline; ondemand: model-only baseline (CPUs 0–3). PTY excludes physical UART/MCU. Exact load commands: CLAIMS.md.

> Keep timer, model, software pipeline and physical-fault scopes separate.

9. Run validity and rule versions

Evidence	Stored v1 acceptance	Stricter v2 replay
Cyclictest	30/30	6/30: idle only
E1–E3, PTY	45/45	0/45: RAW monitor times absent
Original E4 entries	17/25	17/25: unchanged

V2 requires whole-window RAW power timestamps, stressor liveness, and clock fits no older than 5 s when mapping MCU time.

MEASURED v1 observations remain reported with their evidence limitation. Replay rejection is missing proof, not proof that every old run malfunctioned. [5] Results inventory.

> A measured number and a fully qualified comparison are different claims.

10. Timer wake-up under contention

MEASURED – v1; v2 accepts idle runs only

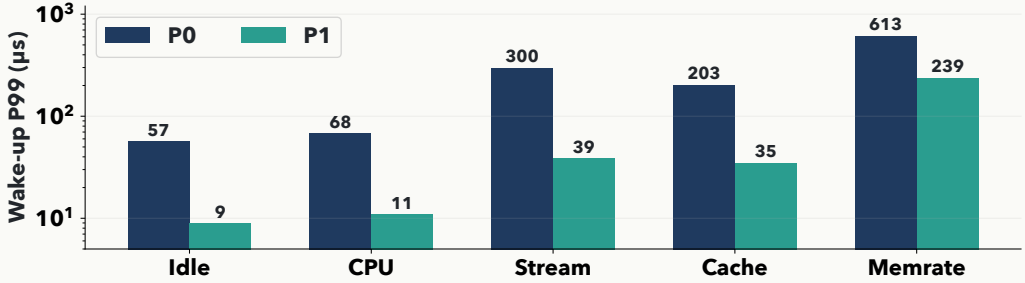


Figure 5: Median run P99; 3 runs/cell, locked memory, CPU 0.

Figure 5: FIFO P99 rises from 9 to 239 μs under memrate.

P0 OTHER / P1 FIFO 50. P50 gap includes timer slack; stressed v2 replay lacks recorded stressor liveness.

Priority reduces CPU tails here; memory-contention tails remain.

11. Inference baseline

MEASURED – accepted subset of a mixed collection

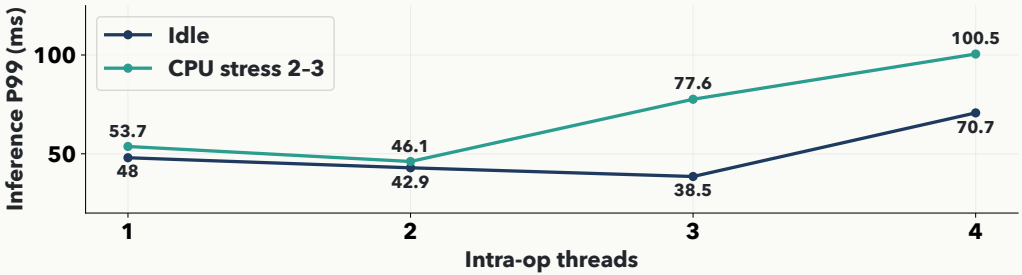


Figure 6: MobileNetV2 FP32, ONNX Runtime; inference only.

Figure 6: idle scaling does not predict loaded tails.

21/24 accepted; idle reps 3/3/3/1, loaded 3/3/3/2. Three 4-thread runs excluded for live under-voltage. Ondemand; affinity CPUs 0–3.

More model threads can compete with the stressed cores.

12. E1: contention reaches the data path

MEASURED – PTY, v1; 15/15 stored, 0/15 under v2 replay

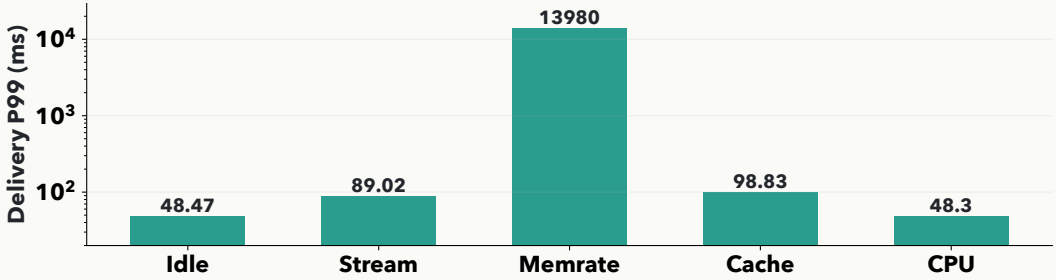


Figure 7: Input-to-Pi-reception P99; median across 3 runs.

Figure 7: memrate reaches 13,980 ms; idle is 48.47 ms.

P0, performance governor. Service includes upstream waiting and pre/postprocessing; this is not model execution time. Logarithmic axis.

The memory-loaded pipeline accumulates seconds of waiting.

13. E2: runtime tuning has limits

MEASURED – PTY, v1; 18/18 stored, 0/18 under v2 replay

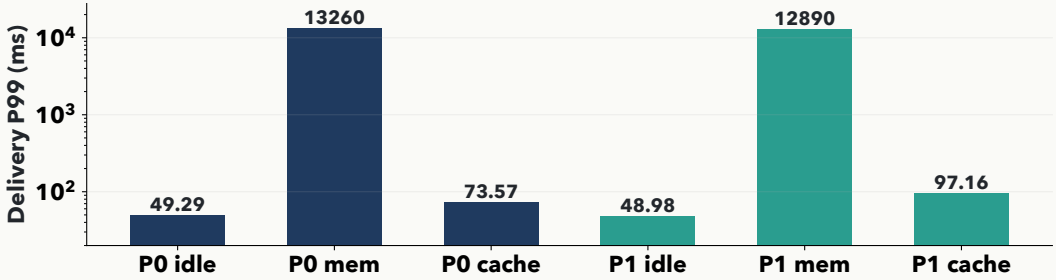


Figure 8: Matched loads; P99 of delivery, median across 3 runs.

Figure 8: memrate P99 stays 13,260 / 12,890 ms for P0 / P1.

P0 source/bridge: CPU 1, unlocked. P1: FIFO 60/70, CPU 1, locked. Both: inference OTHER, CPU 0, unlocked. This compares profiles, not scheduling class alone.

> P1 does not remove memory-loaded backlog or improve every cache result.

14. E3: queue observations and interpretation

MEASURED – PTY, v1; 12/12 stored, 0/12 under v2 replay

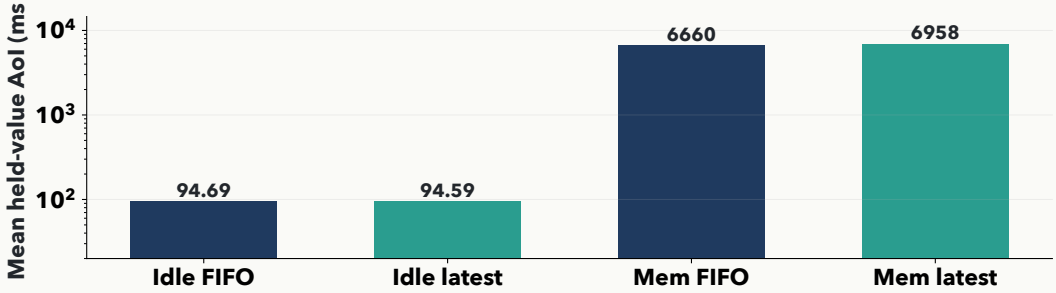


Figure 9: Initialized time-average AoI; stalls and final tail included.

Figure 9: loaded bridge FIFO/latest average ages are 6,660 / 6,958 ms.

MEASURED v1: zero bridge replacements; 100 ms inputs; 122–130 ms completion intervals. FLAGGED E3b: input-latest P99 254–268 ms vs FIFO 13.8–15.9 s; all loaded runs power-invalid.

INFERRED: backlog before inference. Clean input-policy benefit: UNKNOWN.

15. E4: interpret the transition reason

MEASURED – accepted MCU transitions; short-window scopes differ

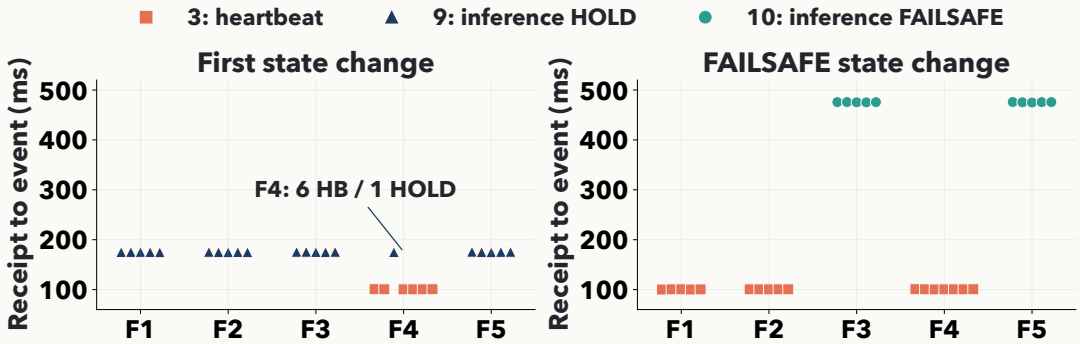


Figure 10: One point per accepted run; horizontal offsets separate close samples.

Figure 10: reason 3 uses heartbeat; reasons 9/10 use inference.

F1 kill, F2 bridge stop, F3 inference stop, F4 FIFO-99 hog, F5 inference suppression. Accepted F1/F2/F3/F5: 5 each; F4: 7/14. F4 first events mix reasons 9 and 3.

About 175/475 ms after receipt matches 200/500 ms source-age thresholds.

16. Independent analyzer cross-check

MEASURED – retained physical captures; collection remains FLAGGED

$\Delta = L - M$; positive means the analyzer interval is longer.
L: last physical UART byte end → both PA5/PB5 low.
M: last accepted heartbeat receipt → FAILSAFE STATE_EVENT time.

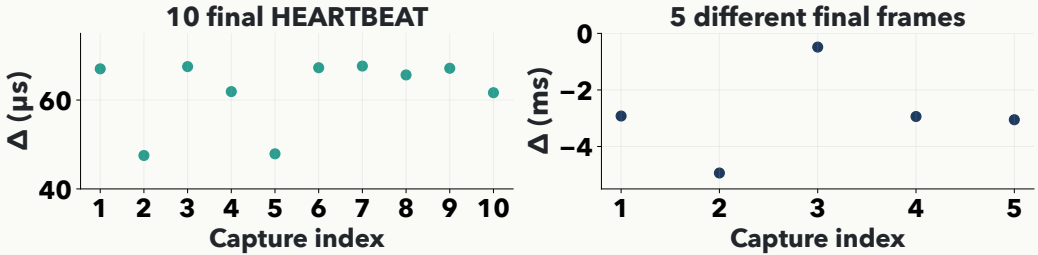


Figure 11: Residuals by capture index; full IDs in qa/LA_capture_key.md.

Figure 11: 10/15 final HEARTBEAT references give +47.5–67.6 μs.

Residual includes timestamp placement, reception/output delay and clock effects. Other 5 end with INFERENCE/ECHO. 8 MHz sampling; absolute calibration UNKNOWN.

Matching frame types aid comparison; they do not qualify the FLAGGED collection.

Re-run 2026-10-20: setup and evidence

MEASURED – new dated evidence; original snapshot retained

Pi 5, 4 GB; Ubuntu 24.04; kernel `6.8.0-1065-raspi`.
Nucleo-F446RE; S1: `st-flash` on Pi, write verified.

Firmware SHA-256 (S1/firmware.sha256):
`bf9a7d956560439704f030237d4b800e`

`df5b131973d2706700be62c3576ff28b`

Analyzer (fx2lafw) and ST-LINK now share the Pi's USB.
8 MHz capture on the system under test, pinned to CPU 3.

Cyclictest: 3×60 s/cell; original: 3×120 s/cell.

Original dates: timer/PTY 11 Oct; E4/LA/E3B 12 Oct. New acceptance uses v2; original comparisons retain v1.
Raw source: results/2026-10-20_rerun.

> Changed kernel, duration and capture host limit direct attribution.

Re-run 2026-10-20: cyclictest comparison

MEASURED – median per-run P99 (μs)

Metric	Original	Re-run	Old n/N	New n/N
P0 idle	57	95	3/3	3/3
P0 CPU	68	183	3/3	3/3
P0 stream	300	675	3/3	3/3
P0 cache	203	403	3/3	3/3
P0 memrate	613	1020	3/3	3/3
P1 idle	9	14	3/3	3/3
P1 CPU	11	13	3/3	3/3
P1 stream	39	42	3/3	3/3
P1 cache	35	37	3/3	3/3
P1 memrate	239	196	3/3	3/3

Original 2026-10-11 (v1); re-run 2026-10-20 (v2). P0 OTHER, P1 FIFO 50; 3 independent runs/cell. Shorter windows do not establish better worst-case tails.

FIFO tails remain sensitive to shared memory contention.

Re-run 2026-10-20: E4 comparison

MEASURED – MCU-local FAILSAFE P50 (ms)

Metric	Original	Re-run	Old n/N	New n/N
F1 P50	100.456	100.427	5/5	5/5
F2 P50	100.577	100.293	5/5	5/5
F3 P50	475.754	474.937	5/5	5/5
F4 P50	100.556	100.426	7/14	5/5
F5 P50	475.596	474.459	5/5	5/5

Reason 3 (F1/F2/F4): last heartbeat; reason 10 (F3/F5): last inference.

Re-run F4 first transition: reason 3; old F4 mixes reasons 3/9.

Original 2026-10-12 includes short F4/F5 windows; new full campaign: 25/25 accepted. F4 original 7/14 = 2/5 full + 5/9 extra. Mapped onset timing is separate; full values in compare/E4_mapped.tex and report.

Compare matching reasons and references; these are observed timings.

Re-run 2026-10-20: analyzer comparison

MEASURED – reference cross-check; no absolute calibration

Metric	Original	Re-run	Old n/N	New n/N
Final HEARTBEAT	10	15	15/15	15/15
Delta min (μs)	47.5243	58.6493	15/15	15/15
Delta max (μs)	67.6493	102.649	15/15	15/15

$\Delta =$ (last UART byte end $\rightarrow$ both GPIO low)
$-$ (last accepted HB $\rightarrow$ FAILSAFE event).

8 MHz: nominal sample period 125 ns. Capture on Pi adds USB/CPU activity to the system under test.

Original 2026-10-12: FLAGGED collection, 10/15 matching; re-run 15/15 matching final HEARTBEAT frames. Matching does not establish collection validity. Marker-to-safe ranges are separately tabulated in the report.

The changed acquisition topology prevents attributing the residual shift.

Re-run 2026-10-20: PS1 and pipeline

MEASURED – PS1 real UART; pipeline PTY evidence

Metric	Original	Re-run	Old n/N	New n/N
Bridge sent	–	602	0/0	1/1
MCU status	–	616	0/0	1/1
Echo accepted	–	62	0/0	1/1
Done-to-send P50 (μs)	–	21.833	0/0	1/1
Done-to-send P99 (μs)	–	42.278	0/0	1/1
Done-to-send max (μs)	–	595.723	0/0	1/1

E1 memrate P99: 10752.7 ms [3/3]
E2 P1 memrate P99: 12890.2 → 8357.15 ms [3/3]
E2 P1 stream P99: 83.2902 ms [3/3]
E3 memrate latest AoI: 2645.45 ms [3/3]
E3B input latest P99: 225 ms [3/3]

E1 ranking changed: memrate + cache became memrate + stream; cache was not run in E2.

E1: memrate vs idle P99; contention reaches delivery: 10752.7 > 45.7999 ms.
E2: memrate P0 vs P1 P99; tuning leaves backlog: 13169.4 > 8357.15 ms.
E3: bridge FIFO vs latest AoI; upstream backlog remains: 3680.48 > 2645.45 ms.

E3B: input FIFO vs latest P99; freshness via replacement: 4698.46 > 225 ms.

PS1 corrected STM32 TX to Pi pin 10; no original counterpart or physical E3 AoI. Pipeline values are per-condition medians, not bounds. Original loaded E3B had no valid runs; new evidence is separate.

Retained pipeline evidence: E1, E2, E3, E3B.

17. Limitations

- 5 V / 3 A supply; live under-voltage excludes loaded runs.
- Original real-UART E3: 18/18 FLAGGED; no clean physical AoI.
- V2 replay exposes missing monitor timestamps and stressor proof.
- Small samples and observed maxima are not worst-case bounds.
- Firmware review fixes compile; they are **not re-flashed or rig-verified**.

Analyzer captures add a GPIO observation to the older stage report; they do not close transit, oscillator, rearm or watchdog qualification.

> The evidence supports this experimental envelope, with explicit exclusions.

18. Conclusion and contributions

- **Timing:** priority improves observed timer tails; shared contention persists.
- **Freshness:** upstream backlog is INFERRED from measured queue behavior.
- **Supervision:** independent MCU timers separate stale work from silence.
- **Artifact:** shared protocol, instrumented pipeline, raw data and exclusions.

Clean input-policy benefit remains **UNKNOWN**; loaded E3b is **FLAGGED**.

The teaching result connects scheduling, IPC, clocks and real-time state decisions.

> Defensible measurements include the failures and the limits of each claim.

19. Future work

PLANNED

- Restore clean power; rerun input-policy and physical E3 under v2.
- Flash reviewed firmware; verify DMA, stack margins and event logging.
- Match analyzer references; calibrate clocks, transit and rearm margins.
- Extend fault exposure and recovery; reach five valid repeats per cell.

Quota and packaged P3 kernel comparisons remain optional matched experiments.

> The next deliverable is qualified physical freshness evidence.

20. References

1. Linux man-pages. *sched(7)*: scheduling policies and affinity.
2. Linux man-pages. *clock_gettime(2)*: monotonic and raw clocks.
3. D. Mills. *RFC 4330*, January 2006, section 5: four-timestamp math.
4. FreeRTOS. *Tasks and Co-routines*: kernel task model.
5. Project artifact. *STAGE_REPORT.md*, *results/README.md*, retained data (11–12 October 2026); source snapshot `5e3f461`.

Primary documentation checked 13 October 2026. Exact figure inputs and hashes: qa/source_hashes.json. Slide-to-evidence mapping: CLAIMS.md.

Backup

Questions, evidence and configuration detail

B1. UART protocol frame

Field	SOF	Version	Type	Seq	Length	Payload + CRC
Bytes	2	1	1	2	1	0–64 + 2

- SOF: A5 5A; little-endian fields; version 3 encoder.
- CRC-16/CCITT-FALSE checks version through payload.
- INFERENCE: 35 bytes, including age-at-send.
- HEARTBEAT carries session identity; STATE_EVENT carries reason.

CALCULATED: 35 bytes × 10 bits / 921600 baud = 0.380 ms nominal line time. This excludes all software delay. Source: protocol/PROTOCOL.md.

Integrity, sequence and freshness checks serve different purposes.

B2. Clock math and measurement boundaries

T_1, T_4: Pi; T_2, T_3: MCU, in microseconds.

$$O = \frac{(T_2 - T_1) + (T_3 - T_4)}{2}, \qquad d = (T_4 - T_1) - (T_3 - T_2)$$

$$O(t) = O_R + D(t - R), \qquad t_{Pi} = R + \frac{t_{MCU} - R - O_R}{1 + D}$$

- Asymmetric delays bias offset; fit residual is not absolute accuracy.
- V2 rejects fits older than 5 s; accepted RMS ceiling is 200 μs.
- Held-value AoI integrates age over initialized time, including final tail.

[3] RFC 4330, section 5; protocol/CLOCK_SYNC.md. RMS ceiling is a configured acceptance choice.

B3. Threshold rationale and rearm

Timer	Configured	Retained observation
Heartbeat	100 ms	F1/F2 reason 3: 100.128–100.641 ms
Source HOLD	200 ms	F5 reason 9: 174.950–175.917 ms after receipt
Source FAILSAFE	500 ms	F5 reason 10: 474.950–475.917 ms after receipt
Healthy rearm	1,000 ms	Core supports it; board port requires reset
Transit allowance	5 ms	Provisional; no validated upper bound

Result age also includes age-at-send: receipt elapsed is not source age.

Configured values are ASSUMED choices, not calibrated safety budgets. main.c; supervisor.c; raw F5 events. E4 threshold snapshots are empty/null.

Healthy heartbeat alone cannot clear latched FAILSAFE.

B4. Full run-validity inventory

Collection	Accepted	Scope / qualification
Cyclictest	30/30 v1	v2: 6/30, idle only
Inference baseline	21/24	Mixed; three live under-voltage exclusions
E1 / E2 / E3	15 / 18 / 12	v1 PTY; v2: 0/45
E3b input policy	3/12	All six loaded runs FLAGGED
Real-UART E3 (original / rerun)	0/18 each	All FLAGGED; distinct validity rules
Original E4	17/25 entries	20 performed; 5 F5 unperformed
Extra F4 / later F5	5/9 / 5/5	Short-window transition checks
Logic analyzer	15 captures	FLAGGED collection; 1 incomplete attempt

Source: results/README.md. Missing STATE_EVENT alone cannot prove unhealthy steady state. No excluded observation is promoted into a clean comparison.

B5. Full E4 table: MCU-local timing

MEASURED – milliseconds; nearest-rank P50 / P95 / max

Fault	Valid	First P50 / P95 / max	FAILSAFE P50 / P95 / max
F1	5/5	175.317 / 175.535 / 175.535	100.456 / 100.641 / 100.641
F2	5/5	175.206 / 175.709 / 175.709	100.577 / 100.596 / 100.596
F3	5/5	175.754 / 175.815 / 175.815	475.754 / 475.815 / 475.815
F4	7/14	100.639 / 175.229 / 175.229	100.556 / 100.931 / 100.931
F5	5/5	175.596 / 175.917 / 175.917	475.596 / 475.917 / 475.917

F1/F2: first 9, FAILSAFE 3.

F3/F5: first 9, FAILSAFE 10. F4: six direct 3; one first 9.

F4 combines original 2/5 with extra 5/9; short windows differ. Reasons 9/10 reference INFERENCE; reason 3 references HEARTBEAT. P95 = maximum at these sample sizes.

> A mixed-reference first-transition summary is not heartbeat latency.

B6. Full E4 table: mapped fault-onset timing

MEASURED – milliseconds; nearest-rank P50 / P95 / max

Fault	Valid	First P50 / P95 / max	FAILSAFE P50 / P95 / max
F1	5/5	83.342 / 83.561 / 83.561	85.357 / 86.209 / 86.209
F2	5/5	83.151 / 83.715 / 83.715	85.715 / 86.113 / 86.113
F3	5/5	83.646 / 85.914 / 85.914	383.620 / 385.888 / 385.888
F4	7/14	85.625 / 87.871 / 87.871	87.277 / 97.344 / 97.344
F5	5/5	83.608 / 85.563 / 85.563	383.603 / 385.558 / 385.558

Mapped intervals begin at Pi software fault onset, through the echo estimator.

MCU-local intervals begin at the last accepted message selected by reason.

Accepted runs record zero false transitions in their checked pre-fault windows.

Clock residual and timestamp placement limit mapping accuracy. Extra F4 and F5 do not qualify original recovery/long-exposure gates. No false-trip rate or worst-case bound follows.

B7. Claim-to-evidence map

Claim	Evidence label	Retained source
FIFO timer tails differ	MEASURED v1	cyclictest summaries
Memory-loaded backlog	MEASURED v1, PTY	E1/E2 tables, infer.csv
Zero replacements; slow completions	MEASURED v1, PTY	E3 table; infer.csv
Backlog before inference	INFERRED	E3 queue observations
Clean input-latest benefit	UNKNOWN; E3b FLAGGED	Power-invalid E3b logs
Heartbeat / source-age decisions	MEASURED, scoped	E4/F5 STATE_EVENT
10/15 matching analyzer references	CALCULATED; collection FLAGGED	VCD + CRC decode + events
Clean physical AoI / safety bound	UNKNOWN	Not established

Analyzer: calculated comparison of retained MEASURED captures; matching frame types do not establish run validity. Full qualifications: CLAIMS.md; hashes: qa/source_hashes.json.

B8. Tools and branch map

Execution environment

Mac: XeLaTeX, plots, cross-compile, flash.

Pi: Linux pipeline, stress-ng, cyclictest, metadata.

STM32: FreeRTOS tasks, TIM2, DMA UART, state GPIO.

Analysis: e_summary, e4_summary, la_analyze.

Artifact provenance

`feat/pi-timing-tools`
`feat/inference-baseline`
`feat/fw-rx-fix`
↓ merged release evidence
`release/os-final`
↓ submission worktree
`feat/os-submission` / `5e3f461`

Branch history is provenance, not a completion claim. This deck build changes only submission/defense; it runs no hardware experiments.

Regenerate this deck with its retained script and source manifest.